

Insights from `synheart_har_pipeline` the 1D-CNN baseline (Synheart Kinematics)

Short notes on the CNN pipeline. This is a much stronger baseline than the starter: it reads the raw signals with a 1D-CNN (no 561 features), uses **LOSO** cross-validation, adds **SO(3) orientation augmentation**, exports to ONNX for on-device use, and keeps all six classes. Credit where due — the orientation instinct is the right one. The notes below are the gaps.

1 The augmentation fights its own posture feature

The three static postures — sitting, standing, lying — carry almost no motion: the body-acceleration motion $\text{std}(\|\mathbf{a}_{\text{body}}\|)$ is $\approx 0.011\text{--}0.016$ g for all three and fully overlapping (Fig. 1, right). The only thing that separates them is **gravity_mean** — the direction of gravity in the phone frame — which alone reaches 85.2% on the three (Fig. 1, left).

But **gravity_mean** is orientation-specific, and the $\text{SO}(3)$ augmentation rotates it over the whole sphere. After a random rotation the same cue drops to 33.6%, i.e. chance ($\approx 34\%$). So the augmentation that buys orientation robustness also **erases the only cue for the static postures**. You cannot have full rotation invariance *and* gravity-based posture from the same input. The notebook presents augmentation as a pure gain; the cost to sitting/standing/lying is never measured.

2 Rotation robustness is never actually tested

The whole pitch is “orientation-robust,” but augmentation is applied only to the *training* set and the model is evaluated on the original waist-fixed test subjects. There is no rotated-test evaluation, so the headline claim is never demonstrated on held-out data.

3 LOSO and the final test number are not comparable

LOSO (30 folds) is run *without* augmentation, while the final reported test accuracy is a single 70/10/20 subject split *with* augmentation. There is no LOSO-with-augmentation, so the real effect of augmentation on a new person is never shown — two different protocols are reported side by side.

4 Heavy for the brief

The model is a $\sim 344\text{K}$ -parameter CNN that needs training, a GPU, and an ONNX runtime. The assignment asks for “very lightweight, no thousands of features, on-device.” A CNN is the heavy end of that scale compared with a few-feature deterministic rule — a real engineering trade-off that is not discussed.

5 Takeaway

The pipeline is on the right track (raw signals, LOSO, augmentation), but it runs into the same wall our work identifies: **motion is orientation-proof, posture is not**. Forcing rotation invariance throws away the gravity direction that the static postures depend on. That is why posture (sitting vs standing vs lying) belongs to a separate gravity-aware stage, not to a fully rotation-invariant model.

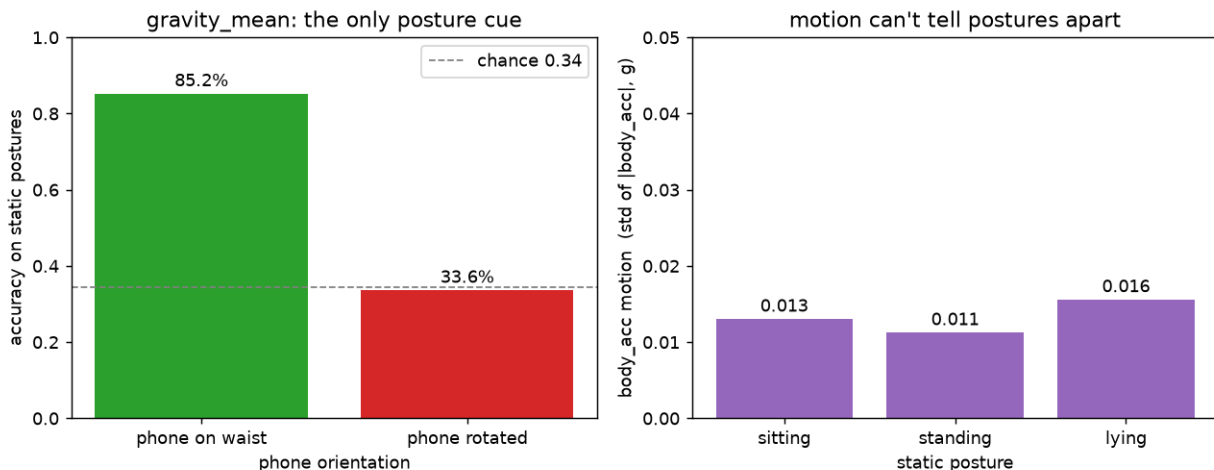


Figure 1: Left: `gravity_mean` separates the three static postures at 85.2% on the waist data, but collapses to chance (33.6%) once the phone is rotated — which is exactly what the $\text{SO}(3)$ augmentation does. Right: body-motion is tiny and overlapping for all three static postures, so motion alone cannot separate them.

6 The other side of the map: the walk types

For completeness, the dynamic classes (walking, ascending, descending) are the mirror image — they are a *motion* question, not a gravity one. Using only orientation-proof motion features from $\|\mathbf{a}_{\text{body}}\|$ (intensity and cadence), the three separate at 55.7% (chance 35.8%): descending stands out (harder impact, faster cadence ≈ 3 Hz) while walking and ascending overlap (Fig. 2). The separation is partial and uses **no gravity at all**. This confirms the division of labour: activity and walk-type live on the orientation-proof motion side, posture lives on the gravity side — and only the latter breaks under rotation.

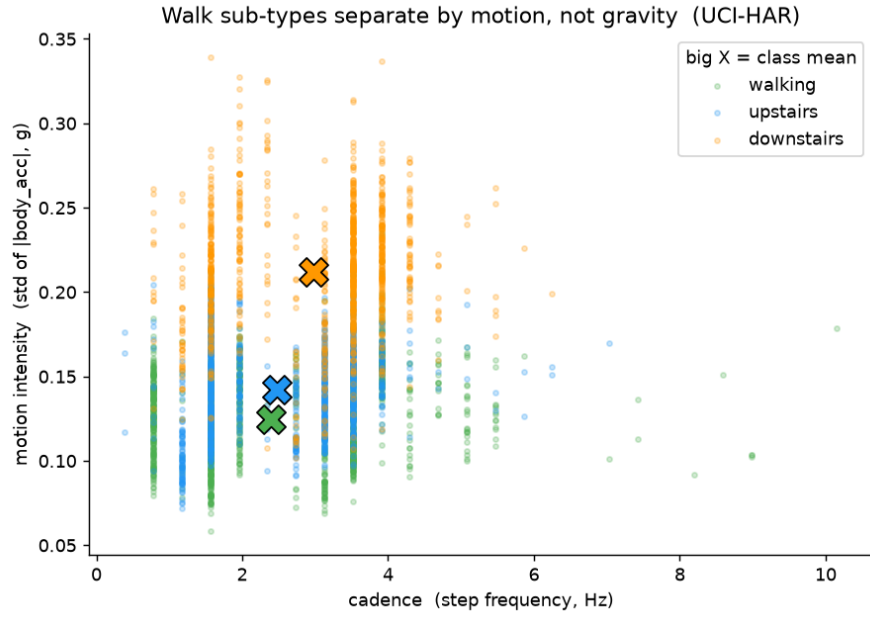


Figure 2: The three walk types in orientation-proof motion space (UCI-HAR). Descending separates by higher intensity and cadence; walking and ascending overlap. No gravity direction is used, so this view is unchanged by phone rotation.